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Group:

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Semester:

4th Semester

Subject:

Database Applications

Lab Task:

SQL Server – DDL Commands

Submitted To:

Engr. Asma Javid

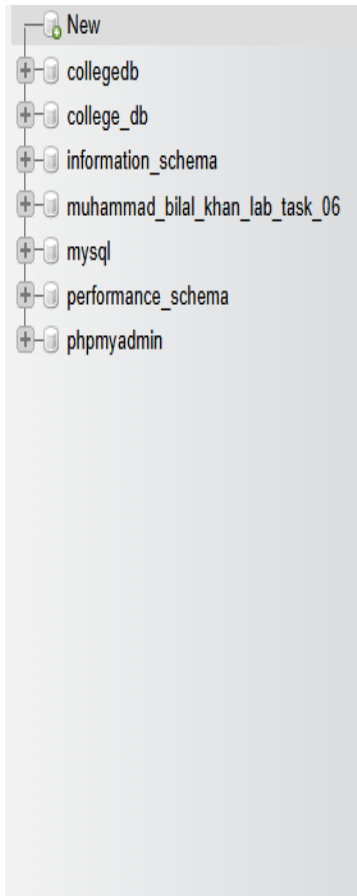
Lab Task 6

Section 1:

Creating a Database and Table Using SQL DDL Commands

1. Create a new database named as your full name in all lowercase with underscores, followed by _lab_task_04.





2. In this database, create a table named Students with the following attributes and appropriate data types:

StudentID (INT, Primary Key)

FirstName (VARCHAR, 50)

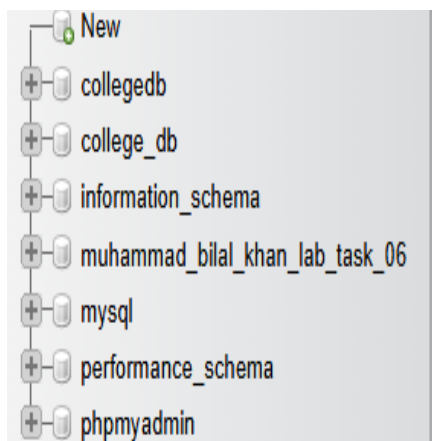
LastName (VARCHAR, 50)

DOB (DATE)

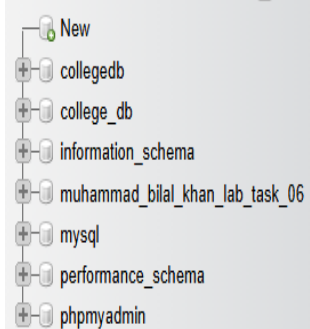
Gender (VARCHAR, 10)

Email (VARCHAR, 100)

PhoneNumber (VARCHAR, 15)



```
1 CREATE TABLE Students
2 (
3 StudentID INT AUTO_INCREMENT PRIMARY KEY,
4 FirstName VARCHAR(50) NOT NULL,
5 LastName VARCHAR(50) NOT NULL,
6 DOB DATE,
7 GENDER VARCHAR(10) NOT NULL,
8 Email VARCHAR(100) NOT NULL,
9 PhoneNumber VARCHAR(15)
10 );
```



✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
USE muhammad_bilal_khan_lab_task_06;
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

⚠ Error: #1046 No database selected

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0176 seconds.)

```
CREATE TABLE Students ( StudentID INT AUTO_INCREMENT PRIMARY KEY, FirstName VARCHAR(50) NOT NULL, LastName VARCHAR(50) NOT NULL, DOB
DATE, GENDER VARCHAR(10) NOT NULL, Email VARCHAR(100) NOT NULL, PhoneNumber VARCHAR(15) );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

The screenshot shows a database management interface. On the left, a tree view displays the database structure, including a 'students' table under a 'muhammad_bilal_khan_lab_task_06' schema. The main panel on the right shows the SQL query `SELECT * FROM `students`` and its results. The results are displayed in a table with the following columns: StudentID, FirstName, LastName, DOB, GENDER, Email, and PhoneNumber. Below the table, there are buttons for 'Query results operations' and 'Create view'.

StudentID	FirstName	LastName	DOB	GENDER	Email	PhoneNumber
-----------	-----------	----------	-----	--------	-------	-------------

Section 2:

Modifying Table Structure Using SQL DDL Commands

1. Add a new column Address of type VARCHAR(255) to the Students table.

The screenshot shows the same database management interface as before. The SQL query editor now contains the command `1 ALTER TABLE students ADD Address VARCHAR(255);`. The results panel is empty, indicating that the command has been executed successfully. At the bottom of the interface, there are buttons for 'Clear', 'Format', and 'Get auto-saved query'.

Navigation pane showing database structure:

- New
- collegedb
 - New
 - students
- college_db
- information_schema
- muhammad_bilal_khan_lab_task_06
 - New
 - students
- mysql
- performance_schema
- phpmyadmin

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0228 seconds.)

```
ALTER TABLE students ADD Address VARCHAR(255);
```

[Edit inline] [Edit] [Create PHP code]

Navigation pane showing database structure:

- New
- collegedb
 - New
 - students
- college_db
- information_schema
- muhammad_bilal_khan_lab_task_06
 - New
 - students
- mysql
- performance_schema
- phpmyadmin

```
SELECT * FROM `students`
```

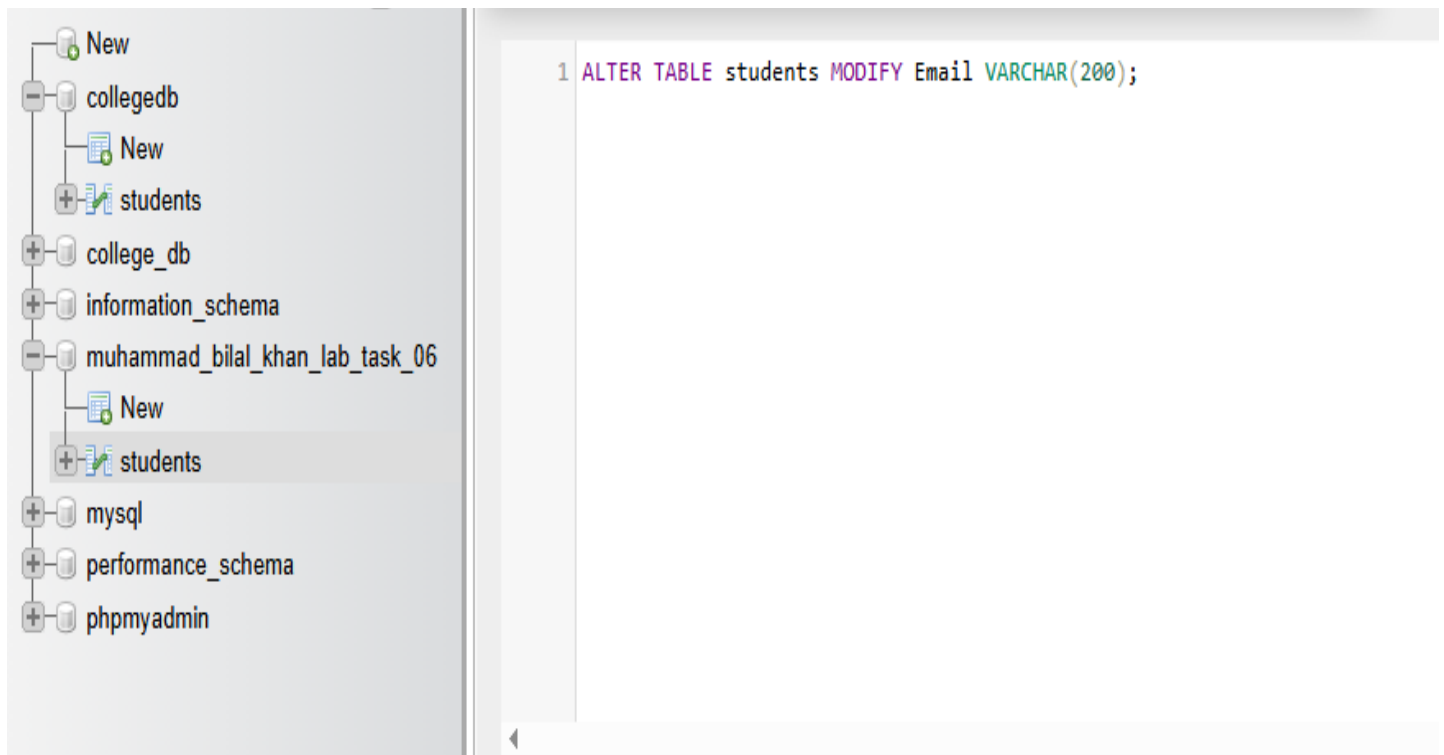
☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

StudentID	FirstName	LastName	DOB	GENDER	Email	PhoneNumber	Address
-----------	-----------	----------	-----	--------	-------	-------------	---------

Query results operations

 Create view

2. Modify the column Email to be of type VARCHAR(200).



The screenshot shows a database management interface. On the left sidebar, the tree view includes 'New', 'collegedb', 'college_db', 'information_schema', 'muhammad_bilal_khan_lab_task_06', 'mysql', 'performance_schema', and 'phpmyadmin'. Under 'muhammad_bilal_khan_lab_task_06', there is a 'New' folder containing the 'students' table, which is currently selected. The main query editor on the right contains the following SQL statement:

```
1 ALTER TABLE students MODIFY Email VARCHAR(200);
```



The screenshot shows the same database management interface as above. The 'students' table is still selected in the sidebar. The main area now displays a success message in a green bar: "✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0542 seconds.)". Below this, the executed SQL query is shown: `ALTER TABLE students MODIFY Email VARCHAR(200);`. At the bottom of the query area, there are three links: "[Edit inline]", "[Edit]", and "[Create PHP code]".

Recent
Favorites

New
collegedb
college_db
information_schema
muhammad_bilal_khan_lab_task_06
New
students
mysql
performance_schema
phpmyadmin

Table structure
Relation view

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	StudentID	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	FirstName	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	LastName	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐ 4	DOB	date			Yes	NULL			Change Drop More
☐ 5	GENDER	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
☐ 6	Email	varchar(200)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 7	PhoneNumber	varchar(15)	utf8mb4_general_ci		Yes	NULL			Change Drop More

☐ Check all
With selected:
 Browse
 Change
 Drop
 Primary
 Unique
 Index
 Spatial

 Fulltext

3. Remove the column PhoneNumber from the Students table

New
college_db
information_schema
muhammad_bilal_khan_lab_task
New
coursedetails
mysql
performance_schema
phpmyadmin

```

1 ALTER TABLE students DROP PhoneNumber;

```


The screenshot shows a database management interface. On the left is a tree view of the database structure:

- college_db
 - information_schema
 - muhammad_bilal_khan_lab_task_06
 - New
 - students
 - mysql
 - performance_schema
 - phpmyadmin

On the right, there is a toolbar with the following options: ☐ Profiling, [\[Edit inline \]](#), [\[Edit \]](#), [\[Explain SQL \]](#), [\[Create PHP code \]](#), and [\[Refresh \]](#).

Below the toolbar is a table header with the following columns: StudentID, FirstName, LastName, DOB, GENDER, Email, and Address.

Below the table header is a section titled "Query results operations" with a button labeled "Create view".

Section 3:

Create the Second Table

1. Create a new table named Courses with the following attributes:

CourseID (INT, Primary Key)

CourseName (VARCHAR, 100)

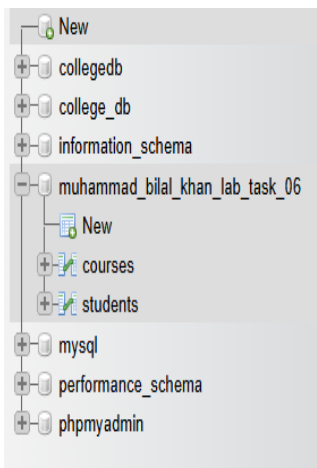
Instructor (VARCHAR, 50)

Credits (INT)

StartDate (DATE)

EndDate (DATE)

Department (VARCHAR, 50)



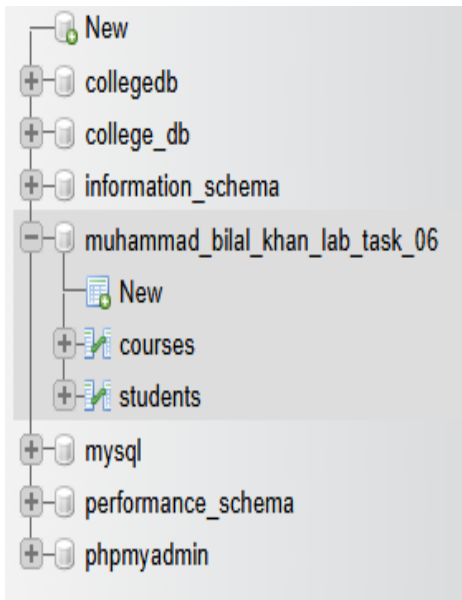
✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0337 seconds.)

```
CREATE TABLE Courses ( CourseID INT PRIMARY KEY, CourseName VARCHAR(100) NOT NULL, Instructor VARCHAR(50) NOT NULL, Credits INT, StartDate DATE, EndDate DATE, Department VARCHAR(50) NOT NULL );
```

[Edit inline] [Edit] [Create PHP code]

Section 4:

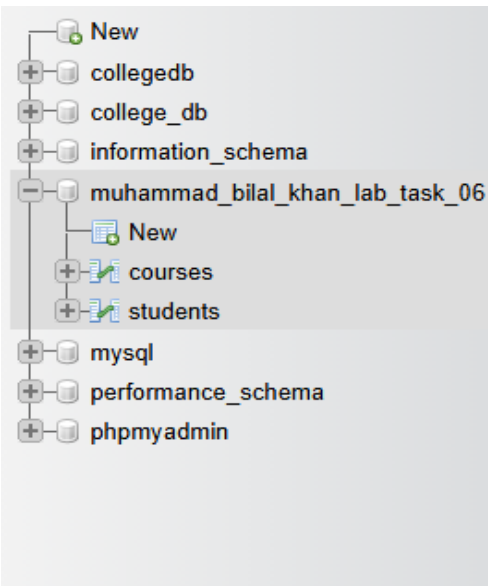
Alter the Second Table



✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0278 seconds.)

```
ALTER TABLE courses ADD CourseDescription TEXT;
```

[Edit inline] [Edit] [Create PHP code]



```
1 ALTER TABLE courses MODIFY Credits DECIMAL(3,1);
```

New

collegedb

college_db

information_schema

muhammad_bilal_khan_lab_task_06

New

courses

students

mysql

performance_schema

phpmyadmin

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0247 seconds.)

ALTER TABLE

courses

MODIFY

Credits

DECIMAL(3,1);

[Edit inline]

[Edit]

[Create PHP code]

New

collegedb

college_db

information_schema

muhammad_bilal_khan_lab_task_06

New

courses

students

mysql

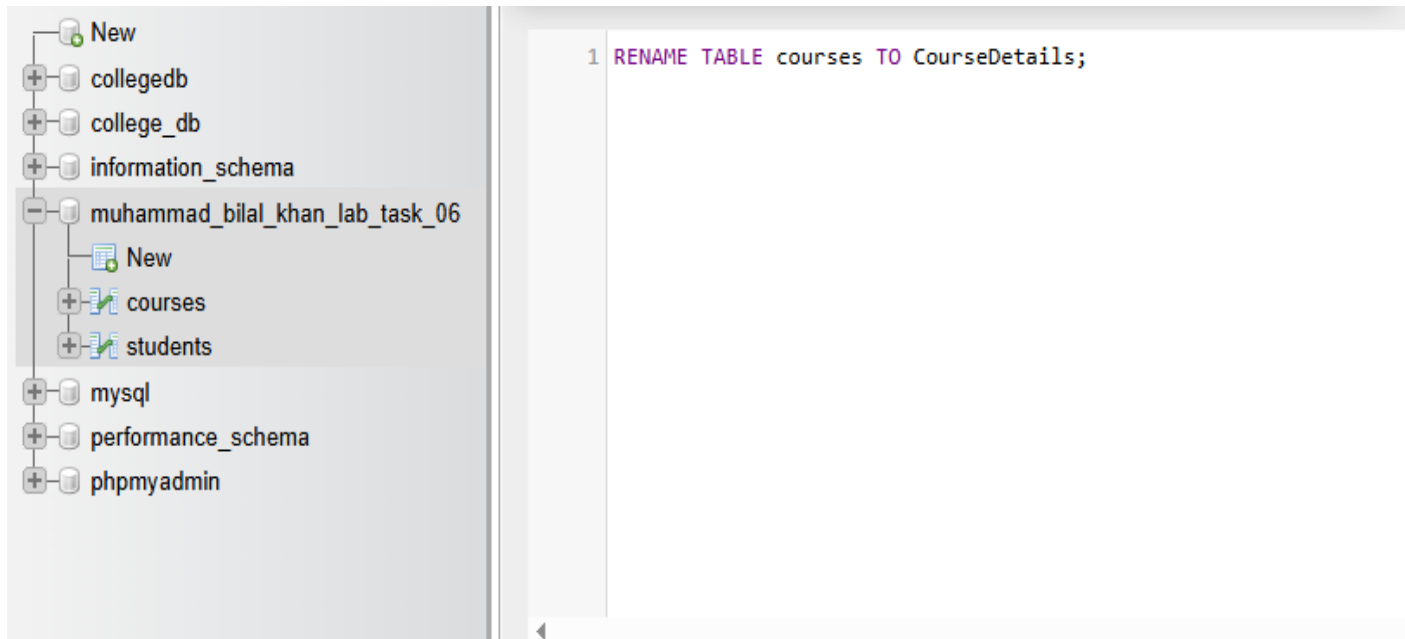
performance_schema

phpmyadmin

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	CourseID 	int(11)			No	None			 Change  Drop  More
☐ 2	CourseName	varchar(100)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐ 3	Instructor	varchar(50)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐ 4	Credits	decimal(3,1)			Yes	NULL			 Change  Drop  More
☐ 5	StartDate	date			Yes	NULL			 Change  Drop  More
☐ 6	EndDate	date			Yes	NULL			 Change  Drop  More
☐ 7	Department	varchar(50)	utf8mb4_general_ci		No	None			 Change  Drop  More
☐ 8	CourseDescription	text	utf8mb4_general_ci		Yes	NULL			 Change  Drop  More

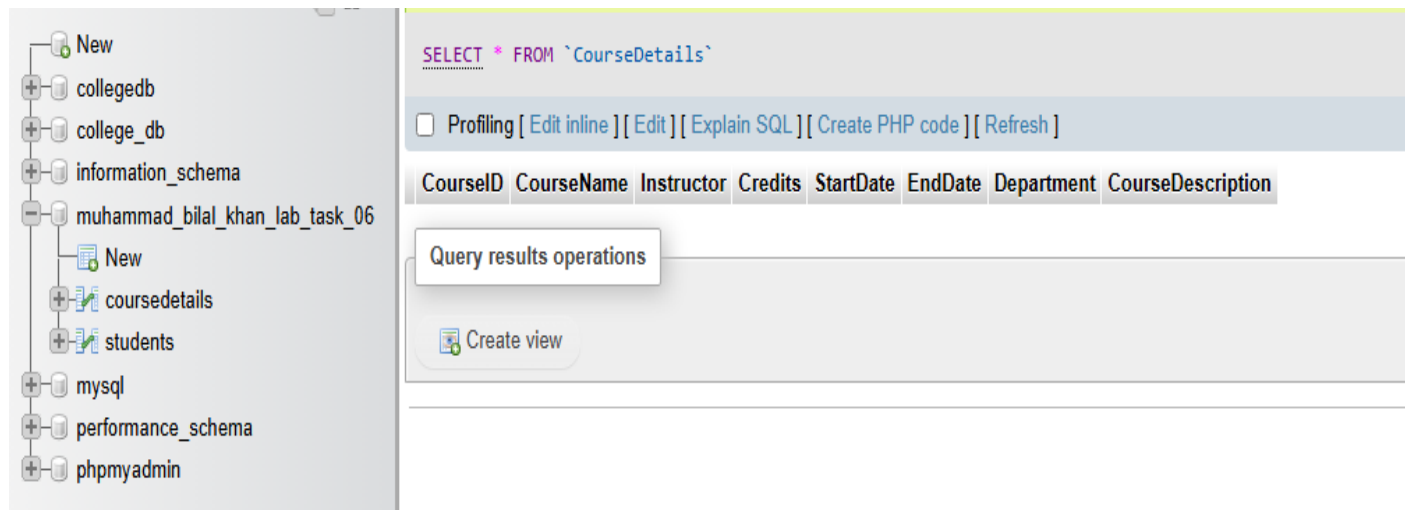
Section 5:

Rename and Delete Tables and Database



The screenshot shows the MySQL Workbench interface. On the left, the 'Schemas' pane displays a tree structure of databases. The database 'muhammad_bilal_khan_lab_task_06' is selected and expanded, showing its contents: 'New', 'courses', and 'students'. The 'courses' table is highlighted. On the right, the 'SQL Editor' pane contains the following SQL statement:

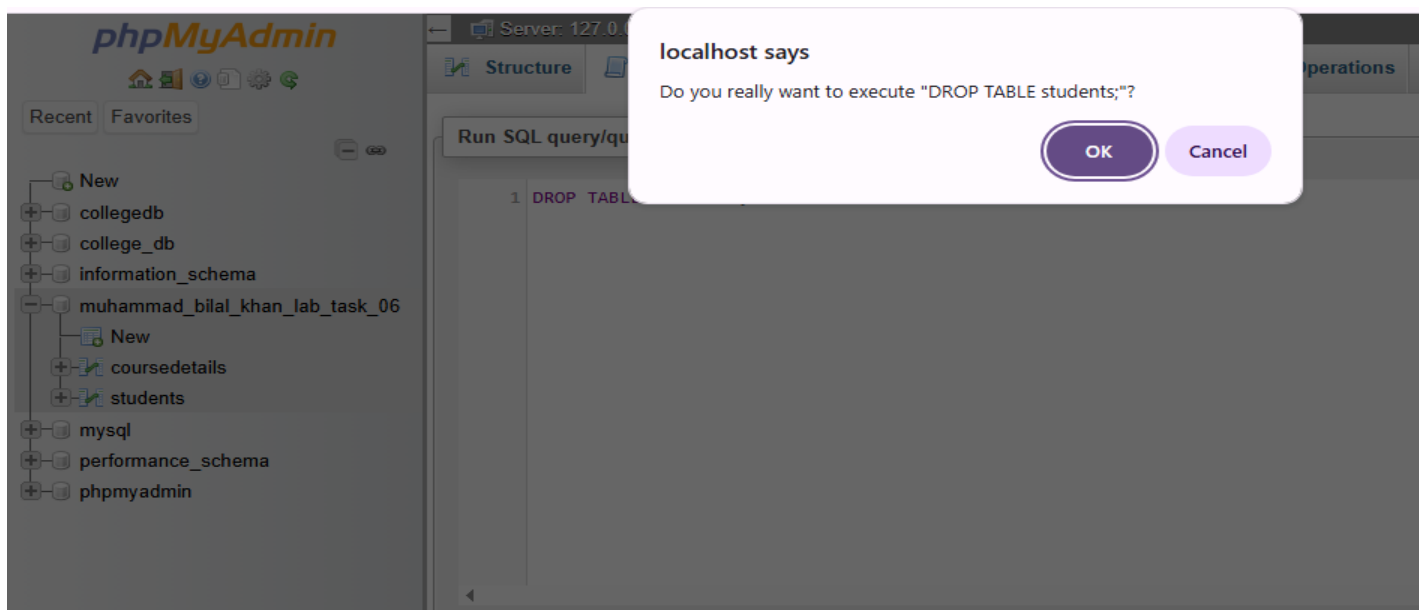
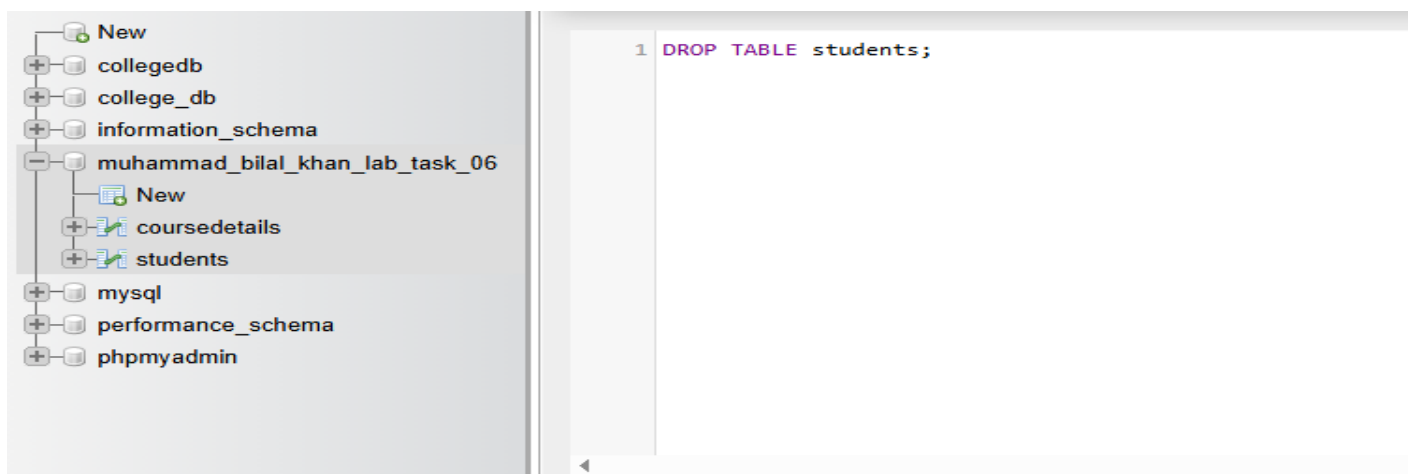
```
1 RENAME TABLE courses TO CourseDetails;
```

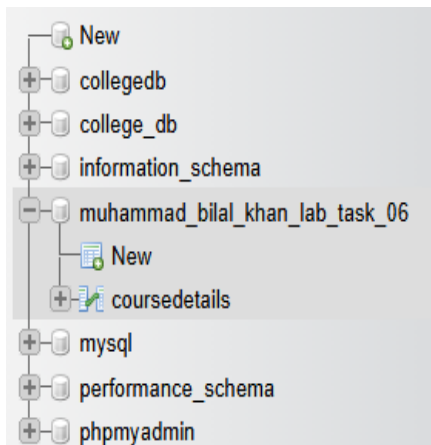


The screenshot shows the MySQL Workbench interface after executing the previous query. On the left, the 'Schemas' pane shows the database 'muhammad_bilal_khan_lab_task_06' expanded, with the table 'coursedetails' now visible. On the right, the 'SQL Editor' pane shows the query:

```
SELECT * FROM `CourseDetails`
```

Below the query, there are several buttons: ☐ Profiling, [\[Edit inline \]](#), [\[Edit \]](#), [\[Explain SQL \]](#), [\[Create PHP code \]](#), and [\[Refresh \]](#). Below these buttons, the query results are displayed in a table with the following columns: CourseID, CourseName, Instructor, Credits, StartDate, EndDate, Department, and CourseDescription. Below the table, there is a section titled 'Query results operations' with a button [\[Create view \]](#).

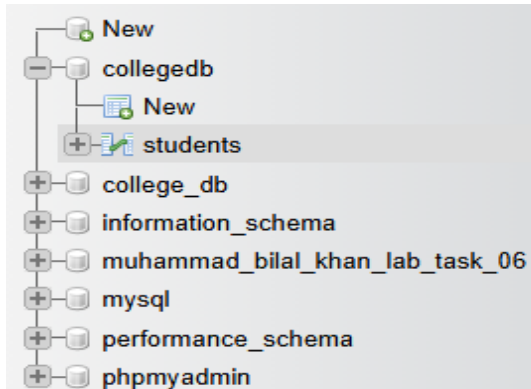




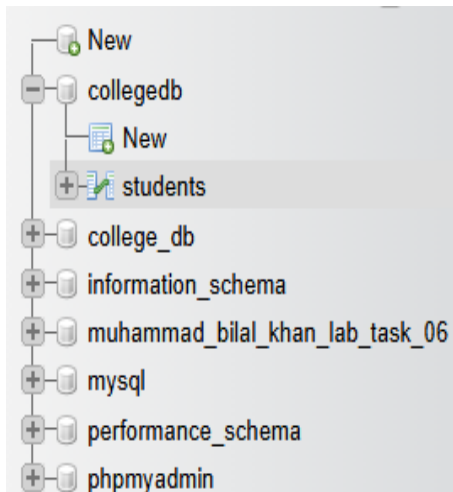
✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0276 seconds.)

```
DROP TABLE students;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]



```
1 USE collegedb;  
2  
3 DROP DATABASE collegedb;
```



✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
USE collegedb;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0316 seconds.)

```
DROP DATABASE collegedb;
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

New

+

college_db

+

information_schema

+

muhammad_bilal_khan_lab_task_06

+

mysql

+

performance_schema

+

phpmyadmin

Create database

Database name

utf8mb4_general_ci

Create

☐ Check all

Drop

Database	Collation	Action
☐ college_db	utf8mb4_general_ci	Check privileges
☐ information_schema	utf8_general_ci	Check privileges
☐ muhammad_bilal_khan_lab_task_06	utf8mb4_general_ci	Check privileges
☐ mysql	utf8mb4_general_ci	Check privileges
☐ performance_schema	utf8_general_ci	Check privileges
☐ phpmyadmin	utf8_bin	Check privileges

Total: 6